Research Statement

Farzaneh Sabbagh • Ph.D. Candidate in Economics, University of New Hampshire • farzaneh.sabbaghziarani@unh.edu

I am an applied microeconomist working at the intersection of environmental economics and local public finance. My research asks a single question from two directions: how do people reveal what public goods are worth to them, and what determines whether communities are willing to pay for those goods? On the demand side, I use original surveys and non-market valuation to measure how recreational users respond to changes in environmental quality. On the supply side, I use administrative election data to study how the demographic makeup of a community shapes its willingness to finance public investment. Together, the two strands span an unusually wide range of empirical approaches: original survey design, joint revealed- and stated-preference valuation, panel and discrete-choice methods, and causal inference. Both strands emphasize careful data construction, transparent identification, and results that a policymaker could actually use.

Job Market Paper: Are Nutrient Policy Impacts on Recreation in Lake Erie as Murky as the Water?

Lake Erie supports a regional economy of more than $50 billion a year, and its recurring harmful algal blooms have prompted the United States and Canada to commit, under the Great Lakes Water Quality Agreement, to cutting phosphorus loading by 40 percent. A cut of that size will change water quality, aquatic species, and most likely bloom frequency, with consequences for recreation, tourism, commercial fishing, and agriculture. My job market paper, joint with Stephanie Brockmann and supported by an external research grant from the Great Lakes Fishery Commission, an independent binational U.S.–Canada agency, takes up the recreation part of that question: recreation is a major regional industry, and its beachgoers and anglers stand to be affected differently and perhaps disproportionately, which the surveys described next were designed to measure.

The empirical core is two contingent-behavior surveys that I designed, fielded, and analyzed: a survey of 1,921 beachgoers drawn from a Qualtrics panel of residents in the counties bordering the lake, and a survey of 467 anglers. Each survey collects respondents' actual trips (their revealed preference, or RP) as well as the number of trips they would make under experimentally varied conditions of bloom extent, odor, bacterial contamination, and fish abundance (their stated preference, or SP). Following Cameron (1992) and Adamowicz et al. (1994), I pool each respondent's RP and SP trip counts and estimate a single negative binomial trip-demand model on the combined data. The model includes a respondent-level random intercept, because the same person contributes several observations; treats bloom severity as a categorical attribute; and includes a stated-preference indicator that lets hypothetical answers sit at a different level than actual behavior, correcting for hypothetical bias, while sharing the same slope coefficients. The RP observations anchor the model in real behavior and identify the travel-cost response. The SP observations supply variation in conditions the lake has not yet experienced. The travel-cost coefficient then converts predicted changes in trips into consumer surplus.

The final step is what sets the paper apart. Rather than valuing arbitrary "improvement" scenarios, I use the estimated demand functions to predict how trips change under the ecological conditions that Ecopath with Ecosim (Christensen and Walters, 2004; Zhang et al., 2023), a calibrated food-web model of Lake Erie, produces for the 40 percent reduction. The model, which tracks how nutrient loading works its way through algae, prey fish, and the predator fish anglers target, was run by a team of ecologists; my contribution is incorporating its outputs into the demand models. Earlier work has fed fish-population model output into recreation demand (Massey et al., 2006; Melstrom et al., 2015) or paired RP and SP data with expert-elicited

scenarios (Ready et al., 2018); to my knowledge, none has run an ecosystem model's simulations through a pooled RP and SP demand model for two user groups. Three results follow. First, the two groups share one response: harmful bacteria are the strongest deterrent for beachgoers and anglers alike, cutting beach trips by roughly a third in every basin and standing as the only bloom characteristic that significantly reduces fishing trips. This matters for the policy because the nutrient target acts on the size of a bloom, and a smaller bloom is not guaranteed to be a symptom-free one, since odor and toxin levels track algal biomass poorly. The paper therefore evaluates every scenario at two bounds, a best case in which the policy also removes odor and bacteria and a worst case in which they remain and argues that monitoring and mitigating bacterial contamination deserves priority alongside load reduction. Second, the groups differ in how they respond to the blooms themselves. Western Basin beachgoers exhibit a threshold response: moderate and severe blooms deter trips almost identically, so a 40 percent reduction that moves the basin from severe to moderate raises trips by only 1.4 percent, whereas an 80 percent reduction that restores mild conditions raises them by 20 percent. For anglers, the spatial extent of a bloom is essentially irrelevant; what matters is the fish. Third, the same nutrient reduction that improves beaches works against anglers. With less phosphorus the lake produces less algae, so prey fish have less to eat and there are fewer of the predator fish anglers target; fish biomass falls and predicted angler trips fall by 30.5 percent. Per person, the resulting angler loss is two to twelve times the average beachgoer's gain, depending on whether the policy also removes the bloom's symptoms. Whether the policy is a net gain for the lake as a whole therefore depends on how many more beachgoers than anglers use it, which the surveys do not measure; the paper reports the break-even instead, roughly two to twelve beachgoers per licensed angler, and because beachgoers vastly outnumber licensed anglers on Lake Erie, aggregate net welfare is plausibly positive. The paper has been presented at the ASSA, Eastern, and Southern Economic Association meetings and is being prepared for journal submission.

My second line of work moves from the demand for a public good to the collective decision to supply one.

How Racial Composition Shapes Local Investment in Education

School bonds are the main way American communities finance long-lived K–12 capital, and they are one of the few high-stakes public finance decisions made directly by voters. The taxpayers who vote are, on average, older and more likely to be non-Hispanic and white than the students who benefit. A large literature documents that racial and ethnic fragmentation lowers support for public goods, but the school-bond literature has measured fragmentation almost exclusively through the population shares of the 'community' overall. In work joint with Robert Mohr and Karen Conway, I ask a sharper question: does the mismatch between the racial and ethnic composition of voters and that of students matter over and above composition itself?

To address this question, I develop a model of voting behavior that is based on Vigdor's (2004) model of collective action and its two-population extension in Kurtz et al. (2024). In this model, a voter's decision is influenced by both their own race or ethnicity and the racial and ethnic composition of the district's student population. The model yields the measure of voter–student divergence, which Kurtz et al (2024) call the Hybrid Fragmentation Index (HFI). The HFI is highest when the voting-age population is least representative of the student population.

I next build a district-by-year panel of Texas school-district bond elections that links Texas Bond Review Board election records to citizen-voting-age-population shares from the ACS (allocated to districts through tract-to-district crosswalks and validated against a land-area alternative), Texas Education Agency student enrollment and economic-disadvantage data, adopted tax rates, and charter-school exposure. Using the HFI as the measure of voter–student divergence, I estimate linear probability and logit models with district and year fixed effects, clustered by district. Voter–student fragmentation is negatively and significantly

associated with bond passage. Because the specification controls for the racial and ethnic shares of the voting-age population, the HFI coefficient captures mismatch over and above voter composition. Replacing the HFI with conventional fragmentation measures computed separately for adults or students yields statistically insignificant estimates, while the estimated effect of the HFI is significant and 1.3 to 2 times larger in magnitude. Allowing the influence of the over-65 electorate to trend over time strengthens rather than weakens the result and reveals a reversal that earlier studies missed: the effect of a district's over-65 population on bond passage is negative early in the sample but becomes significantly more positive over time, suggesting that racial and ethnic mismatch, rather than age itself, may be the real barrier to educational investment. The main result survives numerous checks: alternative race-share controls, population versus land-area weighting, single-election-district exclusions, charter school proximity and campus-charter interactions, and vote share as an alternative outcome.

Environmental Justice and the National Priorities List

With Robert Mohr I wrote the entry on the National Priorities List for *The SAGE Encyclopedia of Environmental Justice* (2026). The chapter synthesizes the evidence on who lives near listed sites, on whether siting is causal or compositional, and on how cleanup timing and remedy choice vary with community characteristics. Writing it sharpened a theme that runs through both of my projects: area-level correlations between demographics and outcomes are easy to find and hard to interpret, and credible designs require attention to the spatial unit at which decisions are actually made.

Research Agenda

Findings in my dissertation direct my future research plans. Over the next several years I plan to develop three related projects. The first extends the ecological–economic framework of my job market paper. Because the demand models are estimated as functions of ecological attributes rather than of policy labels, they can be re-run under any scenario an ecosystem model can produce, including climate-driven changes in bloom timing, invasive-species shocks, or alternative phosphorus targets, and the framework transfers directly to other systems facing nutrient-driven blooms, such as coastal estuaries and the other Great Lakes. I am particularly interested in combining the beachgoer and angler models into a joint welfare accounting that policymakers can use to compare targets on a common footing.

The second project carries the school-bond work to the school level and beyond Texas. A pilot using AI-assisted document retrieval shows that the specific campuses financed by a bond can be recovered from public records, which makes it feasible to identify beneficiary campuses at scale. I can then test whether fragmentation measured at the school that receives the money, where voters actually perceive it, predicts outcomes better than the district average, and whether apparent "diversity discounts" in earlier work reflect aggregation bias.

The third project connects my two research strands directly, and the link runs through population turnover. Environmental conditions shape who chooses to live in a community: a cleaned-up Superfund site or a restored waterfront draws new residents, while chronic pollution pushes others away. When the population turns over, the demographic alignment between the adults who pay for local public goods and the students who use them changes too, and my school-bond work shows that this alignment determines whether investment happens. The project therefore asks a two-step question: do Superfund cleanups change the voter–student composition of the affected district, and does that change show up in the district's willingness to pass school bonds? The timing of cleanups provides the shocks, my Texas bond-election panel provides the outcomes, and the HFI measures the mechanism in between. The design needs both toolkits my dissertation built: the valuation work tells me which environmental changes are large enough to move people, and the collective-choice work tells me what those moves do to public investment.